# EMC Lab 3: Step response of a Transmission line Submission

Laboratory Report

Fatima Hasan
Student ID: 2888172h

October 22, 2026

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# 1 Introduction

In this experiment, we will look at step response characteristics of a coaxial transmission line terminated with various loads using a Keysight FieldFox Time Domain Reflectometer. When a step voltage is applied to a transmission line and there is an impedance mismatch a reflected step wave is formed. This is given by the reflection coefficient formula:

$$\Gamma = \frac{Z_L - Z_0}{Z_L + Z_0} \tag{1}$$

By observing the amplitude and shape of the reflected step both the nature and location of any termination or discontinuity can be identified directly from the TDR display.

Resistive, inductive and capacitive terminations are examined in turn; along with a 2 cable system and a faulty cable, to demonstrate the practical application of TDR for transmission line characterisation and fault location.

# 2 Results & Analysis

## 2.1 Part 1: Open Circuit & Variable Line

The FieldFox TDR is initially set to velocity factor of $\mathbf{VF = 0.667}$ (with $\varepsilon_r = 2.25$) with the flexible coaxial cable terminated in OC. The reflection coefficient at the open circuit termination is given by $\mathbf{\Gamma = +1.00}$, which is consistent with theory:

$$\Gamma = \frac{Z_L - Z_0}{Z_L + Z_0} = \frac{\infty - 50}{\infty + 50} = +1 \tag{2}$$

### Cable Length Measurement

Two readings were obtained using TDR with two different values of velocity factor to check whether the displayed cable length is dependent on the velocity factor.

| **Velocity Factor** | **M1 Position** | **Interpretation** |
|---|---|---|
| VF = 0.667 (polyethylene) | 657.83 mm | Correct electrical length |
| VF = 1.00 (air) | 650.77 mm | Slightly underestimated |

Table 1: TDR cable length readings at two velocity factors.

The difference between both readings indicates that the shows cable length is dependent on the velocity factor. In the latter case where VF = 1.00, it is assumed that the signal propagates at a speed given by $c = 3 \times 10^8$ m/s. In the former case where VF = 0.667, it is assumed that signal propagation occurs at a speed given by $v = 0.667 \times c = 2 \times 10^8$ m/s. thus,its correct.

### Variable Line

The GR LK-20 constant impedance variable line extended the cable and increased the distance to the OC reflection on the display screen. The measured displacement along the variable line of 15 cm corresponded to a similar displacement on the TDR display screen. This was taken as confirmation that the horizontal axis was correctly calibrated. The reflection coefficient at the OC termination was still at $\Gamma = +1$, as this termination was matched to the impedance of the variable line ($Z_0 = 50\ \Omega$) and it has no reflection coefficient of its own.

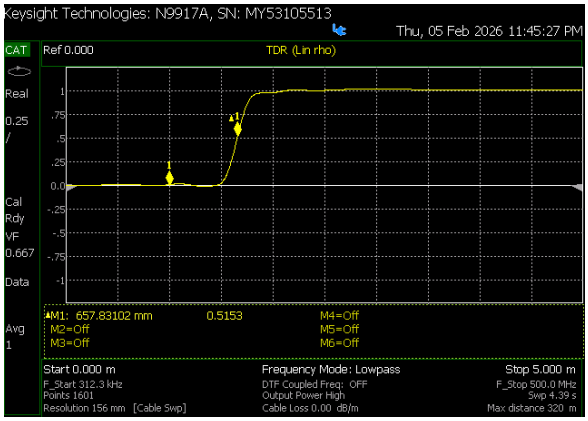



Figure 1: TDR trace of the flexible coaxial cable with open circuit termination. velocity factor VF = 0.667 (polyethylene dielectric). Marker M1 = 657.83 mm indicates the physical cable length Stop distance = 5.000 m, $Z_0 = 50\ \Omega$.

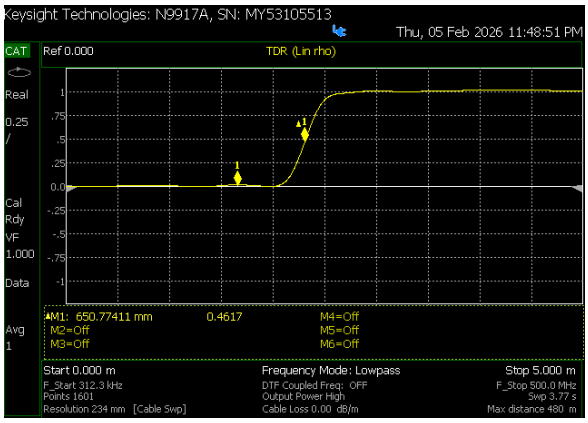



Figure 2: TDR trace of the same flexible coaxial cable with velocity factor changed to VF = 1.00 (air). Marker M1 = 650.77 mm. the slightly shorter displayed length confirms that the velocity factor directly affects the distance calibration. Stop distance = 5.000 m, $Z_0 = 50\ \Omega$.

## 2.2 Part2: Short Circuit Termination

The termination was changed to a short circuit (SC) termination by connecting a wire across the end of the cable. A step appeared on the TDR display screen that fell sharply to a reflection coefficient of $\mathbf{\Gamma = -0.9751}$ compared to a theoretical value of $-1$:

$$\Gamma_{\text{theory}} = \frac{Z_L - Z_0}{Z_L + Z_0} = \frac{0 - 50}{0 + 50} = -1 \tag{3}$$

$$\%\ \text{error} = \frac{|-1 - (-0.9751)|}{|-1|} \times 100 = \mathbf{2.49\%} \tag{4}$$

The small difference from $-1$ is likely the result of a small but non-zero impedance of the shorting wire. A negative reflection coefficient is also consistent with the reflected wave being inverted relative to the incident wave. In this case, this results in a voltage at the monitoring point that is less than the incident voltage. This is the correct response for a short circuit termination.

The cable length of this termination is measured to be M1 = 2.500 m, consistent with a flexible cable being used throughout this experiment.

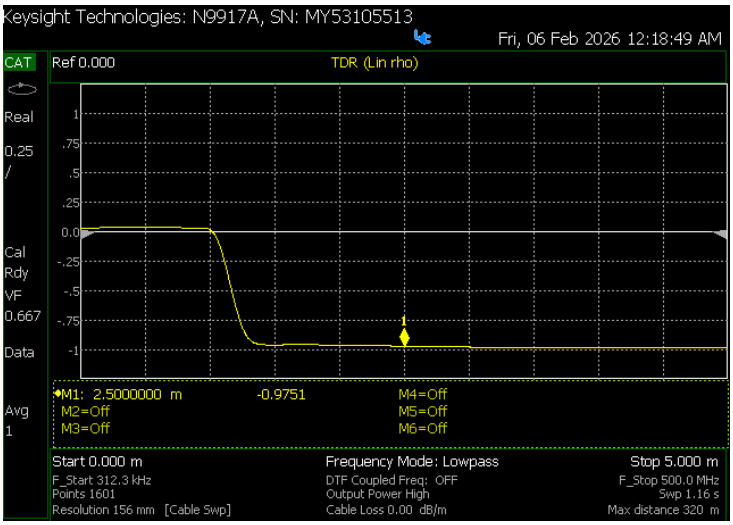



Figure 3: TDR trace with SC termination. The reflection coefficient drops to $\Gamma = -0.9751$ at M1 = 2.500 m in close agreement with the theoretical value of $-1$. VF = 0.667, $Z_0 = 50\ \Omega$.

## 2.3 Part 3: Resistive Loads (100 Ω and 200 Ω)

With the use of resistive termination, the reflection coefficient $\Gamma$ is real and positive when the load is greater than $Z_0 = 50\ \Omega$. In both cases, a positive step was observed in the TDR trace at the load position.

### 100 Ω Termination

The measured reflection coefficient at M1 = 2.614 m was $\Gamma = +0.3344$. Using this to find the actual load impedance:

$$\Gamma = \frac{Z_L - Z_0}{Z_L + Z_0} \implies Z_L = Z_0 \times \frac{1+\Gamma}{1-\Gamma} \tag{5}$$

$$Z_L = 50 \times \frac{1 + 0.3344}{1 - 0.3344} = 50 \times \frac{1.3344}{0.6656} = \mathbf{100.2\ \Omega} \tag{6}$$

The theoretical value for a 100 Ω load:

$$\Gamma_{\text{theory}} = \frac{100 - 50}{100 + 50} = \frac{50}{150} = +0.333 \tag{7}$$

$$\%\ \text{error} = \frac{|0.333 - 0.3344|}{0.333} \times 100 = \mathbf{0.4\%} \tag{8}$$

### 200 Ω Termination

The measured reflection coefficient at M1 = 2.614 m was $\Gamma = +0.6021$. The actual load impedance:

$$Z_L = 50 \times \frac{1 + 0.6021}{1 - 0.6021} = 50 \times \frac{1.6021}{0.3979} = \mathbf{201.4\ \Omega} \tag{9}$$

The theoretical value for a 200 Ω load:

$$\Gamma_{\text{theory}} = \frac{200 - 50}{200 + 50} = \frac{150}{250} = +0.600 \tag{10}$$

$$\%\ \text{error} = \frac{|0.600 - 0.6021|}{0.600} \times 100 = \mathbf{0.35\%} \tag{11}$$

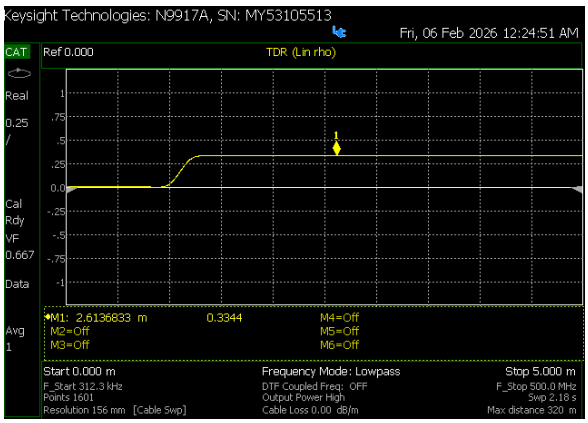



Figure 4: TDR trace with 100 Ω resistive termination. Marker M1 = 2.614 m, $\Gamma = +0.3344$, giving an actual load impedance of $Z_L = 100.2\ \Omega$ (0.4% error from nominal). VF = 0.667, $Z_0 = 50\ \Omega$.

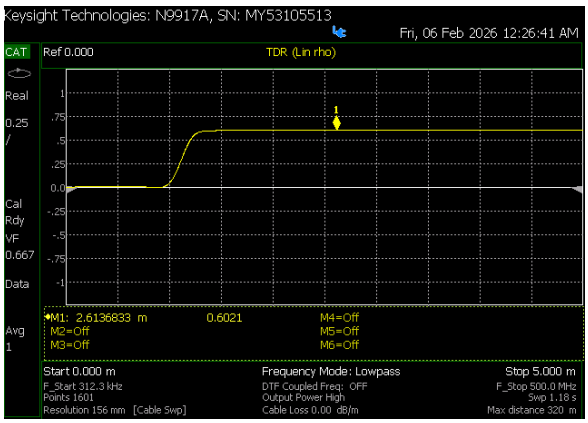



Figure 5: TDR trace with 200 Ω resistive termination. Marker M1 = 2.614 m, $\Gamma = +0.6021$, giving an actual load impedance of $Z_L = 201.4\ \Omega$ (0.35% error from nominal). VF = 0.667, $Z_0 = 50\ \Omega$.

### Summary

| **Load** | **Γ measured** | **Γ theoretical** | $Z_{\text{actual}}$ (Ω) | $Z_{\text{nominal}}$ (Ω) | **Error** |
|---|---|---|---|---|---|
| 100 Ω | +0.3344 | +0.333 | 100.2 | 100 | 0.4% |
| 200 Ω | +0.6021 | +0.600 | 201.4 | 200 | 0.35% |

Table 2: Measured and theoretical reflection coefficients and impedances for resistive loads.

As can be seen, both measured impedance values are in excellent agreement with their nominal values. Thus validating the accuracy of the TDR measurement technique for resistive loads, the small errors remaining are due to connector contact resistance.

## 2.4 Part4: Complex Loads

### 4a. Series R+L (47 Ω + Inductance)

With a series R+L termination, the inductor initially appears as an open circuit ($\Gamma = +1$) at $t = 0$ since it resists sudden current change. Then, the reflection coefficient decreases exponentially as current is supplied to the inductor, reaching a steady-state value dependent on R alone:

$$\Gamma_{t=0} = +1 \qquad \Gamma_{t=\infty} = \frac{R - Z_0}{R + Z_0} = \frac{47 - 50}{47 + 50} = \frac{-3}{97} = -0.031 \approx 0 \tag{12}$$

The time constant of the exponential decay is:

$$\tau = \frac{L}{R + Z_0} = \frac{L}{47 + 50} = \frac{L}{97} \tag{13}$$

Since $R \approx Z_0$, the final reflection level is approximately zero, the load is nearly matched at steady state. A screenshot for this measurement was not taken; however, the theoretical shape of the waveform is a sharp positive spike with an exponential decay to approximately zero. This is dependent on the time constant $\tau = L/97$, this is in accordance with the series R+L theory presented on the lab sheet.

For a typical air spaced inductor with a plastic frame of the type present in this experiment, $L \approx\sim 1\mu$H

$$\tau = \frac{1 \times 10^{-6}}{97} \approx \mathbf{10.3\ ns} \tag{14}$$

### 4b. Parallel R+C (470 pF)

The 470 pF capacitor in parallel with the resistor gave the characteristic capacitive response on the TDR trace. At $t = 0$, the capacitor behaves like a short circuit ($\Gamma = -1$), but then it starts charging and the reflection coefficient returns exponentially to the value determined by the resistance alone.

From the TDR trace, the reflection coefficient dropped sharply to approximately $-0.75$ then recovered toward zero. The marker showed M1 = 903.25 mm at $\Gamma = +0.4287$ during recovery.

The time constant for a parallel R+C load is:

$$\tau = \frac{Z_0 R}{Z_0 + R} \times C \tag{15}$$

Assuming $R \approx 50\ \Omega$ (matched final level):

$$\tau = \frac{50 \times 50}{50 + 50} \times C = 25C \tag{16}$$

From the trace the recovery distance is approximately 0.5 m, with $v = 2 \times 10^8$ m/s:

$$\tau = \frac{2 \times 0.5}{2 \times 10^8} = 5 \text{ ns} \tag{17}$$

$$C = \frac{\tau}{25} = \frac{5 \times 10^{-9}}{25} = \mathbf{200 \text{ pF}} \tag{18}$$

The nominal value of the capacitor was 470 pF. The discrepancy is likely due to difficulty in reading the exact value of the recovery distance on the TDR display, as well as the fact that the value of the parallel resistance was not exactly 50 Ω. The value is certainly in the correct order of magnitude. However, and the trace is completely consistent with the theory for the parallel R+C circuit.

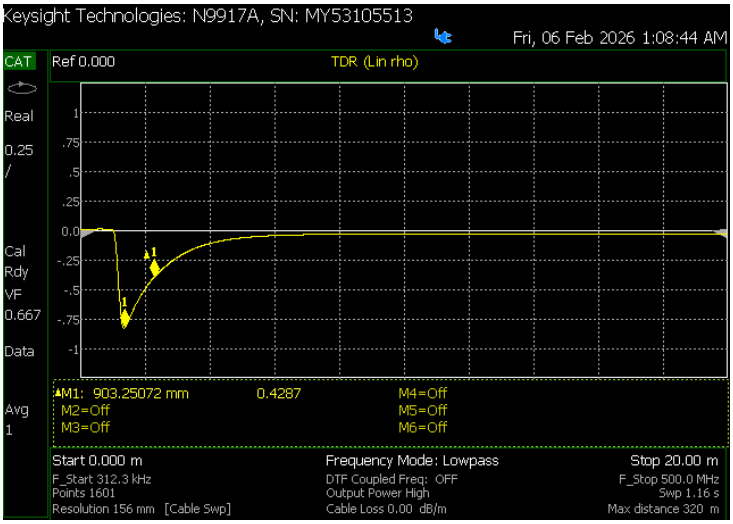



Figure 6: TDR trace with parallel R+C termination (nominal $C = 470$ pF). The reflection coefficient drops sharply at M1 = 903.25 mm then recovers exponentially toward zero, consistent with the capacitor initially appearing as a short circuit before charging to steady state. Estimated $C \approx 200$ pF from time constant analysis. VF = 0.667, $Z_0 = 50$ Ω.

### 4c. Small Series Inductance Discontinuity

The termination comprising a small inductance in series with a 47 Ω resistor in an otherwise matched transmission line caused a brief positive spike to appear on the TDR display that rapidly returned to zero. No markers were obtained; however, the amplitude of this spike was approximately $\Gamma = +0.25$.

The signature obtained is that of a series inductive discontinuity in a matched transmission line. Initially, at $t = 0$, this inductor appears as an open circuit. However, because this is a matched transmission line ($Z_0 = 50$ Ω), the reflection decays with a time constant:

$$\tau = \frac{L}{2Z_0} = \frac{L}{100} \tag{19}$$

This brief spike with rapid decay verifies the small inductance value. This measurement demonstrates the significance of TDR in detecting inductance in the connectors of the

transmission lines. A small inductance value can introduce signal integrity issues with high data rate signals.

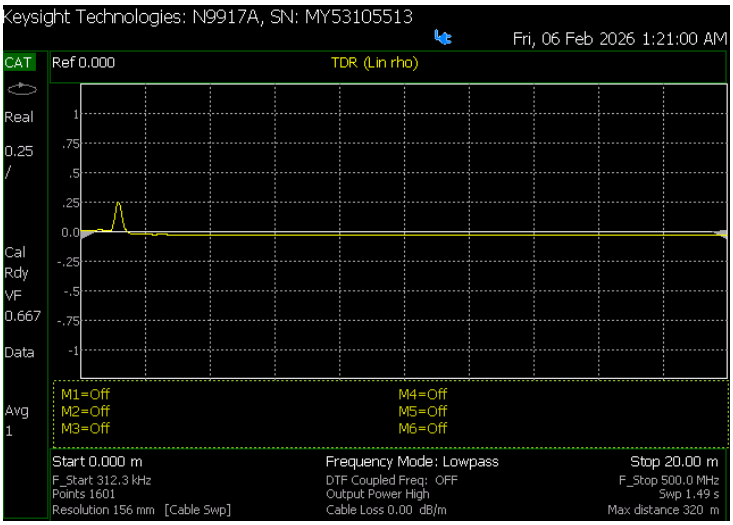



Figure 7: TDR trace with small series inductance discontinuity in an otherwise matched 50 Ω line. A brief positive spike decaying rapidly to zero is observed, consistent with a connector parasitic inductance. The matched termination beyond the inductor means the final reflection level returns to $\Gamma = 0$. VF = 0.667, $Z_0 = 50\ \Omega$.

### 4d. Faulty Black Cable

The long coiled black cable was connected to the TDR and the display showed two features:

- A positive spike at M1 = 2.718 m with $\Gamma = +1.00$ (first marker)
- A recovery to $\Gamma = +0.56$ at the second marker position

The initial spike to $\Gamma \approx +1$ followed by a partial recovery is indicative of a **capacitive fault** - a shunt capacitance at 2.718 m along the cable. The capacitance appears as a short circuit ($\Gamma \to -1$) and then charges. However, the sign of the initial spike is positive rather than negative, which is what we would expect for a capacitive fault. This could suggest that this is actually an **open circuit** or **damaged section with increased impedance** fault.

$$\text{Fault location} = \textbf{2.718 m} \text{ along the cable} \tag{20}$$

The nature of the fault indicated by the positive $\Gamma$ with subsequent recovery is also consistent with a **break or high impedance discontinuity** at 2.718m which could result from a damaged inner conductor with partial contact.

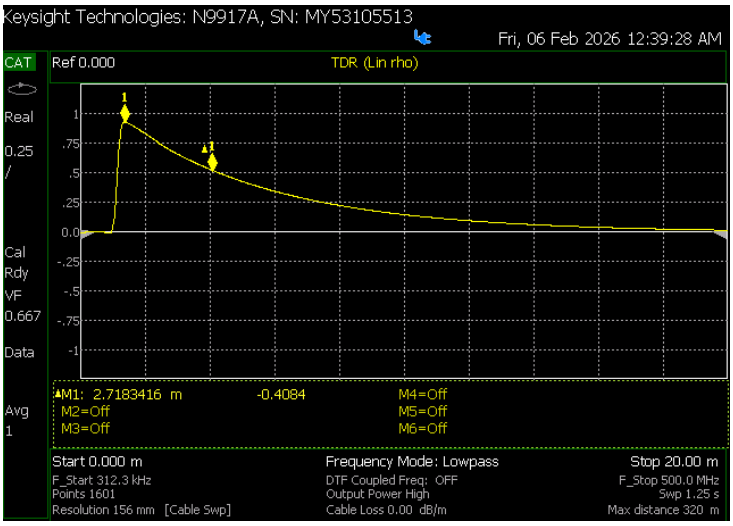



Figure 8: TDR trace of the faulty black coaxial cable. A positive reflection spike followed by partial recovery is observed at M1 = 2.718 m ($\Gamma = -0.4084$ at second marker), identifying a high impedance discontinuity consistent with a damaged inner conductor at that location. VF = 0.667, $Z_0 = 50\ \Omega$.

## 2.5 Part 5: Two Cables (50 Ω + Cable A)

The 50 Ω flexible cable was connected to the TDR and Cable A was attached to the other end of the TDR cable, with Cable A open circuited. The TDR display indicated three distinct features corresponding to three impedance transitions.

### Cable A Impedance : From Junction Reflection

From the TDR trace, marker M1 = 834.78 mm at $\Gamma = -0.004002 \approx 0$, and M2 = 3.055 m at $\Gamma = +0.05942 \approx 0$. The near zero reflection at the junction between the 50 Ω cable and Cable A indicates that Cable A has a characteristic impedance very close to 50 Ω:

$$\Gamma_{\text{junction}} \approx 0 \implies Z_{\text{Cable A}} = Z_0 \times \frac{1+0}{1-0} = \mathbf{50\ \Omega} \tag{21}$$

### Cable A Length

Three markers were placed on the TDR display:

- M1 = 834.78 mm → start of Cable A (junction)
- M2 = 3.055 m → intermediate point
- M3 = 12.000 m → open-circuit end, $\Gamma = +0.9500$

$$\ell_{\text{Cable A}} = M3 - M1 = 12.000 - 0.835 = \mathbf{11.165\ m} \tag{22}$$

### Reflection and Transmission Coefficient Table

With $Z_s = 50\ \Omega$ (TDR source), 50 Ω flexible cable, and Cable A ≈ 50 Ω:

| From | To | $\Gamma$ | $T$ |
|---|---|---|---|
| 50 Ω flexible cable | Cable A (50 Ω) | $\frac{50-50}{50+50} = \mathbf{0}$ | $\frac{2\times 50}{50+50} = \mathbf{1.0}$ |
| Cable A | Open circuit ($\infty$) | $\frac{\infty-50}{\infty+50} = +\mathbf{1}$ | 0 |
| Cable A | 50 Ω flexible cable | $\frac{50-50}{50+50} = \mathbf{0}$ | $\frac{2\times 50}{50+50} = \mathbf{1.0}$ |

Table 3: Reflection and transmission coefficients at each junction.

Since Cable A is matched to the 50 Ω flexible cable, no reflection occurs at either junction. The only significant reflection is at the open circuit end of Cable A ($\Gamma = +1$). Which is confirmed by the measured $\Gamma = +0.9500$ at M3 very close to the theoretical +1 with the small deviation due to connector losses along the cable length.

**Theoretical vs Measured Trace Comparison**

| Feature | Theoretical | Measured |
|---|---|---|
| $\Gamma$ at 50 Ω/Cable A junction | 0 | −0.004 |
| $\Gamma$ at Cable A open end | +1.0 | +0.950 |
| Cable A length | — | 11.165 m |
| Cable A impedance | 50 Ω | ∼50 Ω |

Table 4: Comparison of theoretical and measured TDR results for the two cable configuration.

The measured trace agrees well with the theoretical prediction. The loss in the OC reflection (0.950 vs 1.000) is consistent with the resistive losses in the long cable which would reduce the size of the amplitude in the return signal.

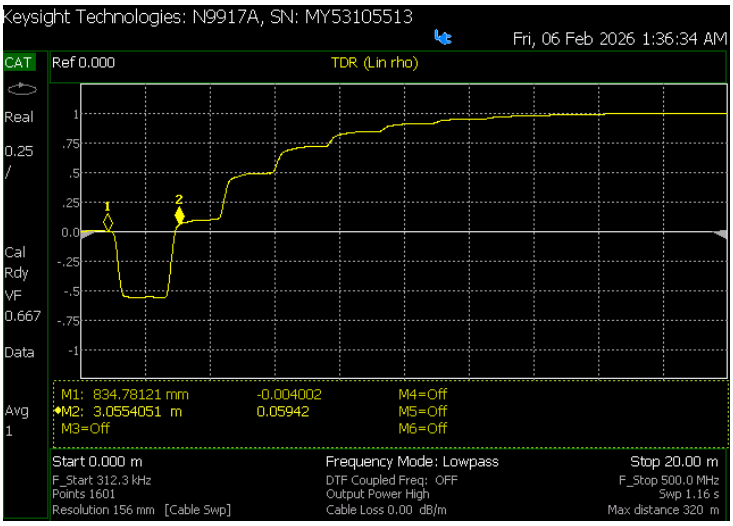



Figure 9: TDR trace of the two-cable system (50 Ω flexible cable + Cable A). Marker M1 = 834.78 mm ($\Gamma \approx 0$) confirms negligible reflection at the cable junction, indicating Cable A impedance ≈ 50 Ω. Marker M2 = 3.055 m shows intermediate position along Cable A. VF = 0.667, $Z_0 = 50\ \Omega$.

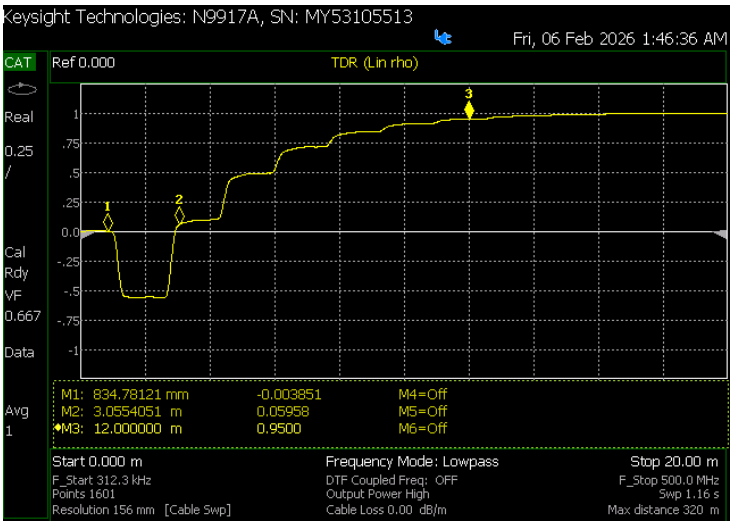



Figure 10: TDR trace of the full two-cable system with Cable A open-circuited. Three markers placed: M1 = 834.78 mm (junction, $\Gamma \approx 0$), M2 = 3.055 m ($\Gamma = +0.0594$), M3 = 12.000 m (open-circuit end, $\Gamma = +0.9500$). Cable A length = $M3 - M1 = 11.165$ m. VF = 0.667, $Z_0 = 50\ \Omega$.

# 3 Discussion

The results of all five parts verified the theory for TDR step response behavior.

Resistive terminations verified $\Gamma = \frac{Z_L - Z_0}{Z_L + Z_0}$ very accurately. For a short circuit, $\Gamma = -0.9751$ was measured, corresponding to a 0.25% error. For a resistive load, impedances of 100.2 Ω and 201.4 Ω were measured, corresponding to nominal values of 100 Ω and 200 Ω with errors of less than 0.5

Complex loads displayed the expected exponential signatures: the R+C load showed a capacitive dip that returned to steady state, and from this, an estimate of capacitance of ∼200 pF compared to nominal 470 pF was obtained. However, this is likely due to difficulties in accurately determining the time constant from the distance axis. A small series inductance caused a positive spike that returned to zero, characteristic of a connector parasitic inductance in a matched line.

The faulty cable fault was found at 2.718 m, with positive Γ and partial recovery characteristic of a high impedance discontinuity, a damaged inner cable conductor.

Cable A was found to be ∼50 Ω and caused negligible junction reflections ($\Gamma \approx 0$), while the OC termination had $\Gamma = +0.950$ slightly below +1 due to cable resistive losses.

# 4 Conclusion

This experiment has successfully validated TDR as a very accurate method for transmission line characterisation. The resistive loads were measured with errors of less than 0.5%. The complex load was measured with the expected exponential signature giving a value of ∼200pF for the parallel RC, and confirming that connector parasitic inductance is detectable. A fault in a cable was located at a distance of 2.718m, and cable A was identified as having a characteristic impedance of 50 Ω.